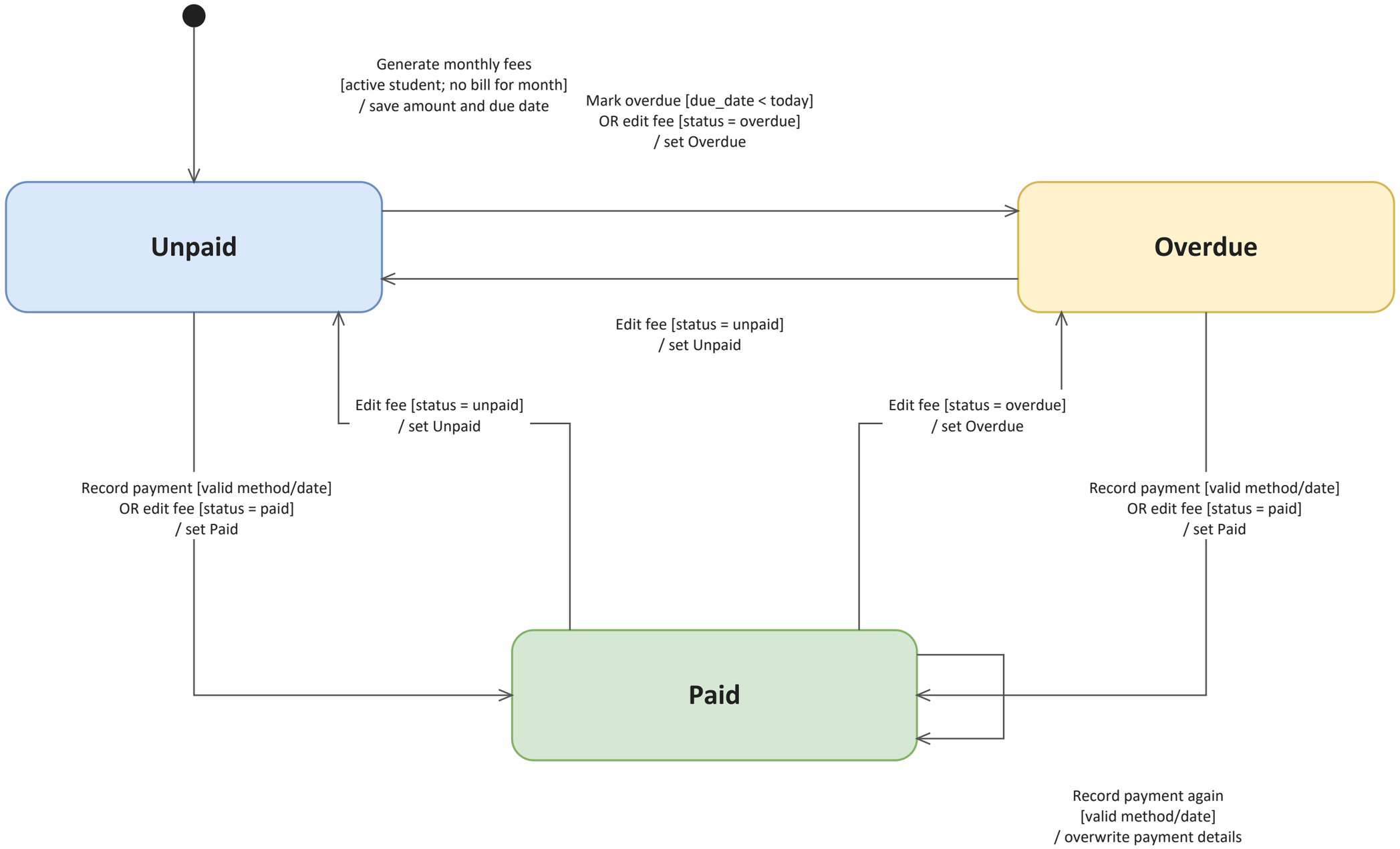
UML State Diagram

Fee lifecycle



State diagram explanation

The state diagram describes the fee lifecycle through Unpaid, Overdue and Paid. The tables explain only the initial node, states and transitions shown in the diagram.

State	Role in the system	Description
Initial node	Starting point of the fee lifecycle.	Generating a monthly fee for an active student without a bill for that month creates an Unpaid fee.
Unpaid	Fee awaiting full payment.	The fee can become Overdue or Paid. Status corrections can also return a fee to this state.
Overdue	Fee recorded as past due.	The fee can become Paid, or a status correction can return it to Unpaid.
Paid	Fee recorded as settled.	The status can be corrected to Unpaid or Overdue. Recording payment again keeps the fee Paid and replaces payment details.

Table. State transitions

From	To	Event and condition	Action
Initial node	Unpaid	Generate monthly fees [active student; no bill for month].	Save amount and due date.
Unpaid	Overdue	Mark overdue [due_date < today] OR edit fee [status = overdue].	Set Overdue.
Overdue	Unpaid	Edit fee [status = unpaid].	Set Unpaid.
Unpaid	Paid	Record payment [valid method/date] OR edit fee [status = paid].	Set Paid.
Overdue	Paid	Record payment [valid method/date] OR edit fee [status = paid].	Set Paid.
Paid	Unpaid	Edit fee [status = unpaid].	Set Unpaid.
Paid	Overdue	Edit fee [status = overdue].	Set Overdue.
Paid	Paid	Record payment again [valid method/date].	Overwrite payment details.